Yashmitha — Dashboard Fix & Upgrade Plan

From: Akilan (Member 4 - ML) | Feed this entire PDF to Antigravity to get step-by-step guidance. Data File: outputs/bowler_spells_dataset.csv | Repo: github.com/therealakilan220/BeyondTheWickets

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PART 1: Fixes to the Current Dashboard (3 Issues)

Fix 1: Chepauk vs Chinnaswamy — Remove All Other Venues

Problem: All 30+ stadiums are showing bars instead of just the two comparison venues.

Steps:

1. Go to the Chepauk vs Chinnaswamy sheet tab (bottom of Tableau).
2. Drag the venue dimension to the Filters shelf.
3. Uncheck all venues. Check ONLY these two:

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MA Chidambaram Stadium, Chepauk
M Chinnaswamy Stadium
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4. Right-click venue color legend → Edit Colors → Set Chepauk = Red, Chinnaswamy = Blue.

Reference numbers to verify: Chepauk Humidity = 72.9%, Dew Spread = 5.87 deg C | Chinnaswamy Humidity = 56.1%, Dew Spread = 11.0 deg C

Fix 2: Grip Loss Heatmap — Replace Venue with Bowler Type in Rows

Problem: Venue is in the Rows shelf so every stadium repeats across all 3 dew zones. It should show Pace vs Spin.

Steps:

1. Go to the Grip Loss Heatmap sheet.
2. Remove venue from the Rows shelf (right-click → Remove).
3. Drag bowler_type to the Rows shelf.
4. Confirm Dew Risk Zone calculated field is on Columns shelf.
5. AVG(economy_rate) should be on Color and Label.
6. Add Filter: innings = 2 (drag innings to Filters → select 2).
7. Change mark type to Square.

Final result should be a 2x3 matrix: Pace and Spin as rows, Dry / Moderate Dew / Severe Dew Risk as columns. Economy Rate should increase left to right as dew worsens.

Fix 3: Add Reference Line at 6 deg C Dew Spread

Problem: No threshold line showing where the critical dew condensation zone begins.

Steps:

1. On any chart that uses dew_spread as an axis, right-click the axis.
2. Click "Add Reference Line".
3. Set Type = Constant, Value = 6.
4. Label it: "Condensation Zone Threshold".
5. Set line color = Red, dashed style.

PART 2: Three New Visualizations to Add (No Pie, Bar, or Line Charts)

Antigravity note: All three charts below use bowler_spells_dataset.csv. Column names to use: dew_spread, economy_rate, bowler_type, venue, boundary_frequency, season_year, innings, pitch_degradation_proxy.

New Chart 1: Scatter / Bubble Plot — Dew Spread vs Economy Rate

Why add it: This is the single most powerful chart for your project. It directly visualizes the inverse correlation between dew spread and economy rate — as dew spread shrinks (more dew), economy rate rises. This validates the core ML finding visually without needing any model.

Shelf	What to Drag

Columns	AVG(dew_spread)
Rows	AVG(economy_rate)
Detail	venue
Color	bowler_type (Pace = Blue, Spin = Orange)
Size	COUNT(Number of Records) — number of spells
Filter	innings = 2, is_night_match = 1

Add a Trend Line: Analytics pane (top right of Tableau) → drag "Trend Line" onto the chart. Add separate trend lines per bowler_type. Add the 6 deg C reference line on the X-axis.

New Chart 2: Box-and-Whisker Plot — Economy Rate Distribution by Dew Zone

Why add it: A box plot shows not just the average but the full distribution (median, spread, outliers). It proves that high-dew matches have a wider, higher distribution of economy rates — showing that dew creates unpredictability for bowlers, not just a flat increase.

Shelf	What to Drag
Columns	Dew Risk Zone (Dry, Moderate, Severe)
Rows	economy_rate
Mark Type	Change to "Circle" in the dropdown
Detail	bowler_type
Filter	innings = 2

After placing, go to Analytics pane → drag "Box Plot" onto the view. Tableau will automatically compute quartiles and draw whiskers.

New Chart 3: Packed Bubble Chart — Venue Economy Rate Weighted by Dew Exposure

Why add it: This gives a premium, visually striking overview of all IPL venues where bubble size shows how many spells were bowled there and color shows average dew severity. High-dew coastal venues (Mumbai, Chennai) stand out immediately as large, dark bubbles.

Shelf	What to Drag
Mark Type	Change to "Circle" in the Mark dropdown
Size	COUNT(Number of Records) — spell count
Color	AVG(dew_spread) — use diverging palette, reverse it so low=red
Label	venue
Detail	venue
Tooltip	AVG(economy_rate), AVG(dew_spread), COUNT spells

In the Show Me panel (top right), click the Packed Bubbles icon to auto-arrange. Annotate Chepauk and Wankhede bubbles to call out the highest dew venues.

Final Dashboard Layout Suggestion

Position	Chart
Top Left	Stadium Dew Map (keep existing, fix filters)
Top Right	NEW: Scatter — Dew Spread vs Economy Rate
Middle Left	Chepauk vs Chinnaswamy (after Fix 1)
Middle Right	NEW: Box-and-Whisker — Economy by Dew Zone
Bottom Left	Grip Loss Heatmap (after Fix 2)
Bottom Right	NEW: Packed Bubbles — Venue Dew Exposure Map